

Table 1: Product Backlog – Infra App

ID	Feature	Story ID	User Story	Priority
1	Dynamic Visual Workflows	1.1	As an engineer, I want to view infrastructure components and their relationships visually to understand system dependencies.	High
2	Control Panel Integration	2.1	As an engineer, I want to deploy and restart services from one interface to save time and reduce manual effort.	High
3	ML Metrics Monitoring	3.1	As a data engineer, I want to track ML model performance and receive alerts on anomalies to maintain accuracy.	Medium
4	GitLab Repository Search	4.1	As a developer, I want to filter repositories quickly and access details within the same dashboard.	Medium
5	User Management & Access Control	5.1	As an admin, I want to manage user roles and permissions to ensure secure access.	Medium

Technology	Learning Curve	Component Ecosystem	Performance	Dashboard Templates	Real-time Updates	Enterprise Adoption
React.js	Moderate	Extensive (Recharts, React Flow, MUI)	Good (Virtual DOM)	Many available	Excellent with hooks	Very High
Vue.js	Easy	Good but smaller	Good	Some available	Good	Growing
Angular	Steep	Good (comprehensive)	Good	Some available	Good	High
Svelte	Easy	Limited	Excellent	Few available	Good	Low

Table 2: Frontend Framework Comparison

Technology	Performance	Auto Documentation	Type Safety	Async Support	Learning Curve	Data Validation
FastAPI	Excellent (async)	Built-in (Swagger)	Pydantic validation	Native	Low	Automatic
Flask	Moderate	Manual setup required	None (manual)	Add-on only	Low	Manual implementation
Django	Moderate	Manual setup required	Limited (forms)	Limited	High	Forms-based
Express.js	Excellent (async)	Manual setup required	Requires TypeScript	Native	Moderate	Manual implementation

Table 3: Backend Framework Comparison